Kojo B. Osei-Ntansah

Aerospace PhD student in the Aerial Robotics and Experimental Autonomy Lab at Georgia Tech — autonomous sUAS, flight controls, and rapid hardware development.

FAA Part 107 Certified Remote Pilot

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EDUCATION

Georgia Institute of Technology Aug 2026 to Present
Ph.D. in Aerospace Engineering · Advised by Dr. Jonathan Rogers, AREAL

Howard University Aug 2022 to May 2026
B.S. in Mechanical Engineering · GPA 3.86 · Summa Cum Laude · Karsh STEM Scholar (full-ride merit scholarship)

EXPERIENCE

Graduate Research Assistant
Aerial Robotics and Experimental Autonomy Lab (AREAL), Georgia Institute of Technology Aug 2026 to Present

- Developing an extended Kalman filter for range-only localization of radio-tagged wildlife from RSSI, with observability-aware flight planning and the airborne SDR receiver that produces the measurement.

Student Researcher
@FTERLab, Applied Fluids and Thermal Engineering Research Laboratory, Howard University Aug 2023 to Jul 2026

- Designed and flight validated 3D printed mounts carrying a 753 g OPC-N3 and iMet-XQ2 payload, standardized across every quadrotor in the lab so datasets were platform agnostic.
- Built and flight tested autonomous atmospheric sampling quadrotors across 50+ flight tests on 5 platforms in 3 years, automating four sampling geometries with concurrent OPC-N3 logging.
- Built flightgen, a Flask web app running on the aircraft's Raspberry Pi that generates, launches, monitors, and aborts autonomous missions across 4 pattern geometries from a browser.

Research Intern, DoW HBCU/MI Summer Research Internship Program
DEVCOM Army Research Laboratory, Manufacturing Science and Technology Branch Jun 2026 to Aug 2026

- Developed rapid-deployment methods for fully 3D-printable 5-inch sUAS quadcopters at DEVCOM ARL; assembly documentation delivered to a U.S. Army infantry division of 10,000+ soldiers cut the build protocol from a full workday to 2-3 hours.
- CAD modeled and 3D printed a modular snap-fit 5-inch quadcopter airframe whose arms and structural parts swap without fasteners or tools.

SURE Research Intern
Aerial Robotics and Experimental Autonomy Lab (AREAL), Georgia Institute of Technology May 2024 to Aug 2024

- Supported adaptive fault tolerant control research for fixed wing elevon failure at Georgia Tech AREAL, and developed a rudder redesign through 3 CAD revisions.

Research Intern, DoD HBCU/MI Summer Research Internship Program
DEVCOM Army Research Laboratory May 2025 to Aug 2025

- Designed and prototyped a lightweight two-axis 3D-printed gimbal for an Arducam optical sensor at DEVCOM ARL, targeting ~90 degrees of motion per axis.

Senior Project Team Lead
Howard University Aug 2025 to May 2026

- Led a 4-person senior design team on an electropulsing fatigue study of grey cast iron; designed the test fixture and specimen geometry and planned the fatigue campaign.

NSF REU Research Intern
Unmanned Aerial Systems Laboratory, Cal Poly Pomona May 2023 to Aug 2023

- Design-build-fly of a tandem-wing eVTOL at Cal Poly Pomona's UAS Lab (NSF REU): OpenVSP/X-Plane

simulation, CAD, fabrication, and flight testing.

PUBLICATIONS

- Osei-Ntansah, K., Reliford, A., and Smith, S. T., 'Development of Autonomous Strategies for High Resolution Data Acquisition for sUAS-Based Atmospheric Research,' AIAA SciTech Forum, 2026.
- Osei-Ntansah, K., Reliford, A., and Smith, S. T., 'Design and Implementation of 3D Printed Sensor Mounts for sUAS-Based Atmospheric Research,' AIAA SciTech Forum, 2025, p. 0192.

SKILLS

Design & Analysis — SolidWorks, Fusion 360, Creo PTC, OpenVSP, ANSYS, FEA, CFD, Structural analysis
Software — Python, MATLAB, Simulink, DroneKit, MAVLink/pymavlink, Flask, HTML/CSS, Arduino, Raspberry Pi, Linux, Git, Monte Carlo simulation
Estimation & Autonomy — Extended Kalman filtering, Particle filters, Range-only localization, Path-loss modeling, Information-driven path planning, Sensor fusion
RF & Sensing — Software-defined radio (Airspy, RTL-SDR), VHF antennas, LNAs and bandpass filters, SNR and link-budget analysis, GPS integration, SensorGnome/Motus pipeline
Flight Systems — ArduPilot, PX4, Cube Orange, Mission Planner, QGroundControl, SITL/HITL simulation, X-Plane, RFD900x/Herelink telemetry
Fabrication & Test — FFF 3D printing, CNC machining, Electronics integration and wiring, Thrust testing, Flight testing (50+ tests across 5 quadrotor platforms)

HONORS AND AWARDS

- GEM PhD Fellowship Awardee (declined), The National GEM Consortium, 2026
- B.S. Mechanical Engineering, Summa Cum Laude, Howard University, 2026
- Karsh STEM Scholar (full-ride merit scholarship), Howard University, 2022
- FOCUS Scholar (2024 and 2025), Georgia Institute of Technology, 2025
- Certificate of Completion, Aerospace Propulsion Course, Massachusetts Institute of Technology, 2025

ACTIVITIES

- **Team Captain, Men's Lacrosse**, Howard University (2023 to 2026)
- **Executive Board, American Society of Mechanical Engineers**, Howard University (2022 to 2023)
- **Member, National Society of Black Engineers**, Howard University (2022 to 2026)
- **Member, Robotics Club**, Howard University (2022 to 2026)